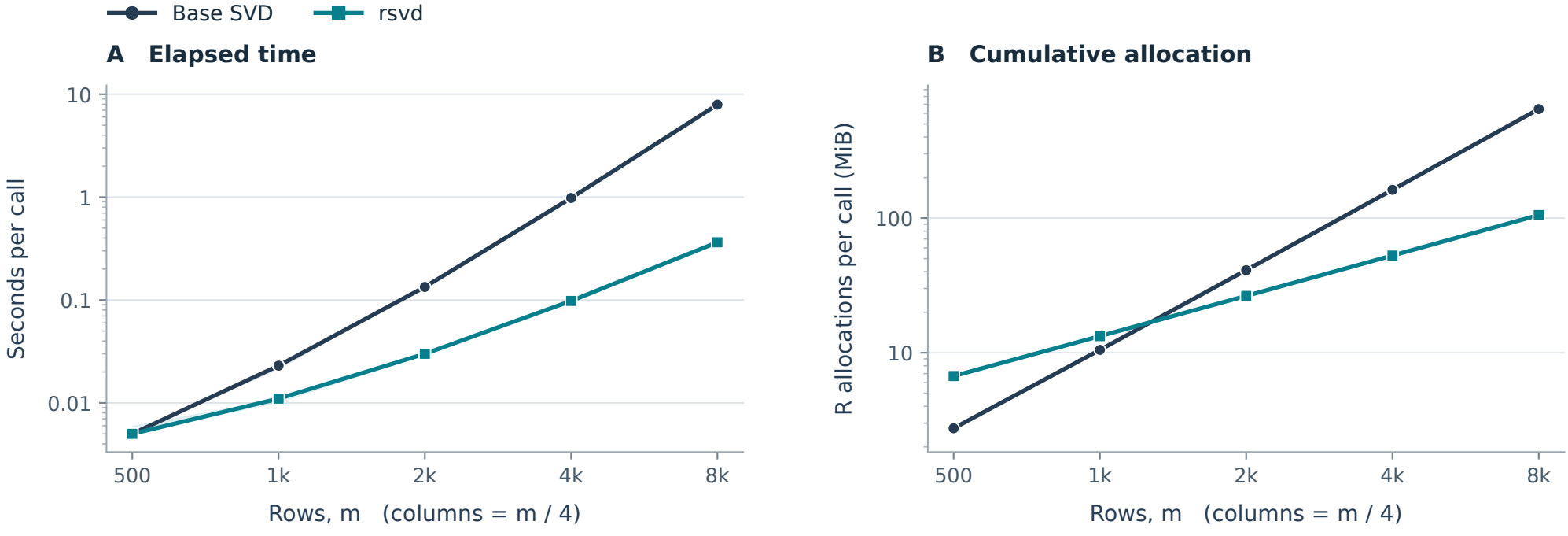
Dense SVD as matrix dimensions grow

Full-rank matrices with a geometric low-rank signal and Gaussian noise



k = 40, p = 10, q = 2 · timing: 5-run medians, IQR shaded · allocations: one profile · OpenBLAS, one thread

Largest observed error / optimal rank-40 error: 1.0006. R allocations exclude setup and do not measure peak RSS.